

Why randomization assignment of treatments matters

Presentation Outline

1 Examples: Random assignment of treatments

- Ames traffic accidents
- Mediterranean diet study retraction

Four types of studies:

Reminder:

Choice of units	Treatment assignment	
	Random	Not
Random	experiment (rare)	survey
Not	experimental study	observational study



Do the distinctions matter: Treatment assignment

- Random assignment $\Rightarrow$

- Not random $\Rightarrow$



Example: Traffic at Ames road intersections

- 3 types of intersections (approximate numbers)
 - Uncontrolled: 0.1 accidents / 10,000 cars / intersection / year
 - Stop signs: 0.6 accidents / 10,000 cars / intersection / year
 - Traffic light: 0.9 accidents / 10,000 cars / intersection / year
- Questions:
 - Do traffic lights “cause” more accidents?
 - Should you request city council to remove the traffic lights?
 - Can you conclude anything from these results??



Example: Med diet study retraction

Study on Mediterranean Diet effects on heart disease

- Published in New England Journal of Medicine
 - Said all participants were individually randomized
- Retracted¹
 - 467 households had 2 participants: got same diet
 - 1 of 11 study centers randomized by clinic, not individuals
- New paper², outlining the deviations, using new statistical analyses

¹doi: 10.1056/ because that was not correct NEJMoa18003891

²doi: 10.1056/NEJMoa1800389



Important points:

Why randomization matters

- Random assignment of treatments to units
 - Make causal claims about the treatments based on just the data and the study design
 - If not randomized, can only make descriptive claims
- Scientists often make generalizations and causal claims
 - If not randomized, those need to be justified by subject-matter knowledge